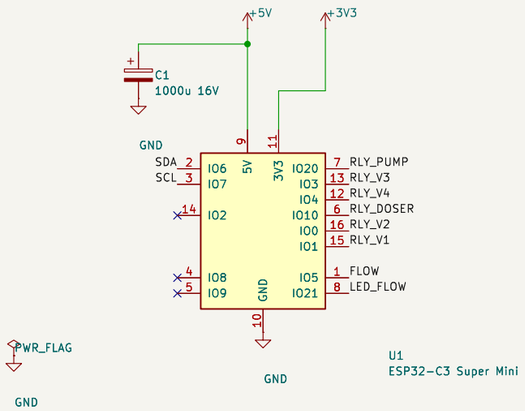
1 MCU AND POWER

ESP32-C3 Super Mini, plugged into 2 x 8-pin sockets (plugs in after the board is screwed down).

USB-C on the module powers the whole board: 5 V / 2 A adapter. 5V pin = USB VBUS.

3V3 pin = the module's LDO (pull-ups, flow input, flow LED). C1 rides out relay switching.



IO8 = on-module blue LED (WiFi status). IO9 = on-module BOOT button (setup after boot).

IO2 = strapping pin, left unused (as Mini-C3).

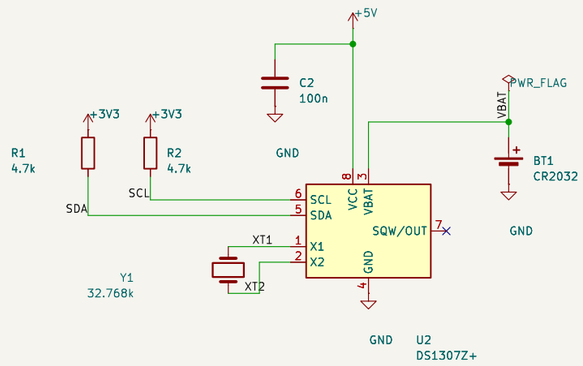
Relay drives: IO20 pump, IO10 doser, IO1 V1, IO0 V2, IO3 V3, IO4 V4.

2 REAL-TIME CLOCK

DS1307 at I2C 0x68, on 5 V (it needs 4.5-5.5 V). Pull-ups to 3V3: it reads 3.3 V as high.

Backup: CR2032 in a vertical holder (measure the holder's pins before ordering).

Crystal 32.768 kHz, 12.5 pF load, through-hole C26 cylinder (as Mini-C3 / FG1), lying flat.

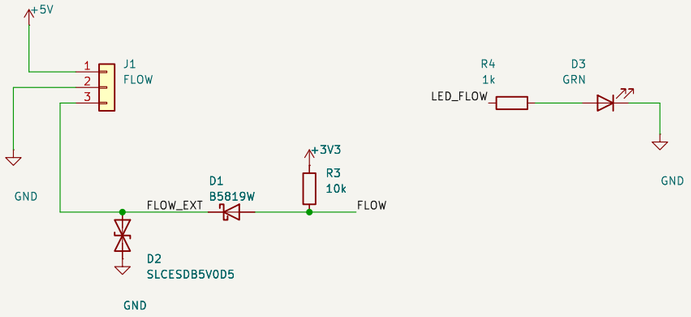


3 FLOW INPUT AND FLOW LED

J1 FLOW: Hall-effect sensor (YF-S201 type) on 5 V. A 2-wire pulse meter uses SIG + GND.

The sensor output swings to 5 V: Schottky D1 only pulls IO5 low, R3 pulls it up to 3V3.

D2: SLCESDB5V0D5 5 V bidirectional TVS (SOD-523, hand-solder pads) on the field cable. FLOW LED on IO21.



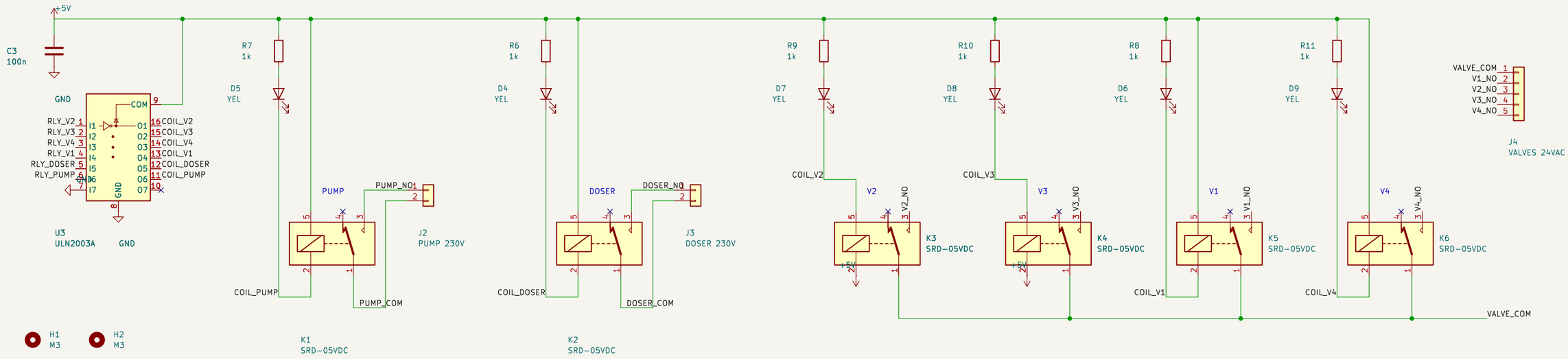
4 RELAY DRIVER AND OUTPUTS

ULN2003A: channels 1-6 straight from the GPIOs (no 74HC595). COM to +5V clamps the coil flyback. Ch 7 unused.

Each relay LED + 1k sits across its coil: lit while the relay is energised.

SRD-05VDC-SL-C: 1 COM, 2/5 coil, 3 NO, 4 NC (unused). K1 PUMP + K2 DOSER: 230 V, >= 5 mm to everything else (.kicad_dru).

K3-K6: 12/24 VAC valves, one COM. K1/K2 NC is live (= COM) while off: the PCB keeps it in the 230 V class.



Mounting: 2 x M3, 92 mm apart on the centre line (same IP65 box bosses as the ESP32 test jig).

99 x 48 mm, 2 per panel; K1/K2 230 V, K3-K6 12/24 VAC

Generated by tools/gen_sch.py from tools/design.py — edit design.py and regenerate

Agri Sensors and Controls

Sheet: /

File: WM-MiniC3.kicad_sch

Title: WM-Mini C3 — pump, doser and 4-valve irrigation controller

Size: A3	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 1/1